Project Based Learning Report on

Incident Analysis and Hotspot Identification for Security Provision against Women Harassment in Public Spaces

MASTER OF COMPUTER APPLICATIONS
OF
VISVESVARAYA TECHNOLOGICAL UNIVERSITY



By

Aman Yadav[1BY25MC012]

Anusha K[1BY25MC017]

II Semester

Under the Guidance of

Dr. M.Sridevi
Faculty, Department of MCA

Department of MCA
BMSITM
Bengaluru-560119



Department of Master of Computer Applications
(Accredited by NBA, New Delhi)

BMS Institute of Technology & Management
(Autonomous Institution under VTU)
Bengaluru – 560119

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PO 3: Design, develop and investigate problems with an innovative approach for solutions incorporating ESG/SDG goals.

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PO 7: Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

PO 8: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

ABSTRACT

Women's sense of safety on urban streets is shaped by lighting, crowd presence, surveillance, and proximity to emergency services, yet these signals are rarely available as a single, actionable map for citizens. This Project Based Learning (PBL) work presents SafeHer, a dual-channel data-collection system for Bengaluru: an OpenStreetMap (OSM)–derived street safety baseline covering approximately 46,424 streets across 53 Bruhat Bengaluru Mahanagara Palike (BBMP) areas, and a live crowdsourcing platform through which women can anonymously mark how a place feels or report an incident with GPS context.

The deliverables include an Expo/React Native mobile application (SafeHer) with Home, Report, Map, and Emergency SOS tabs; a Next.js public website for browser-based reporting, community map, and helplines; a password-protected admin portal for OSM key performance indicators and Generated-versus-Real community monitoring; and a Supabase PostgreSQL schema with row-level security for anonymous inserts and approved-report reads. Street scores use a transparent nine-factor weighted model (police, hospital, bus stops, lighting, CCTV, crowd proxies, commercial activity, night amenity, and road class) classified into Safe, Moderate, and Unsafe bands.

This report documents Phase-1 completion of data collection and platform demonstration. Finished exploratory data analysis, unsupervised hotspot clustering (for example K-Means), and production-grade predictive policing dashboards remain explicit future work and are not claimed as completed outcomes.

Keywords: Women's street safety, OpenStreetMap, Crowdsourcing, SafeHer, BBMP, Supabase, Expo, Next.js, Bengaluru, Anonymous reporting

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1. INTRODUCTION

1.1 Project Description

This PBL project designs and implements SafeHer, a practical platform that combines an offline OSM infrastructure baseline with online citizen reporting for women's street safety in Bengaluru. The OSM pipeline queries Overpass for highway geometries and nearby points of interest inside each of 53 BBMP area bounding boxes, then computes a per-street safety score. The live platform lets a user confirm GPS-derived locality, choose whether the place feels safe or unsafe (or that an incident occurred), optionally attach environmental factors, and store the report anonymously.

Citizens interact through a mobile application (primary channel for GPS and SOS) and a public website (browser fallback without installation). Project staff monitor OSM coverage and incoming community signals through an admin portal. The present phase emphasises trustworthy data collection and usable interfaces rather than final statistical hotspot products.

1.2 Background

Urban safety research consistently shows that risk and fear of crime concentrate in particular places and times rather than spreading evenly across a city. Official crime registers capture only reported events and often lack fine-grained street context. Volunteered geographic information such as OpenStreetMap encodes lighting hints, transport stops, and amenity locations that can serve as environmental proxies. Parallel to this, crowdsourcing platforms demonstrate that residents can contribute experiential safety ratings when privacy and interaction cost are handled carefully.

Bengaluru's BBMP wards provide a natural administrative partitioning for both dataset generation and on-device area labelling. Matching GPS coordinates to a named ward gives users a recognisable place label without exposing a continuous tracking profile.

1.3 Motivation

Three gaps motivated this work:

• Emergency-only apps help after a crisis but do not accumulate place-level safety memory for route planning.

• OSM notebooks alone do not give women a simple way to contribute lived experience back into a shared map.

• Closed civic datasets are hard for student teams to refresh; an open OSM baseline plus anonymous reports is reproducible within a PBL timeline.

SafeHer therefore pairs a transparent scored street dataset with lightweight reporting and SOS tools so that Phase-1 yields both reusable data and a working demonstration for stakeholders.

1.4 Problem Statement

To design and implement a Phase-1 data-collection system that can:

• Extract OSM street and amenity attributes for Bengaluru BBMP areas and publish a scored street safety dataset.

• Enable anonymous GPS-linked safe / unsafe / incident reporting on mobile and web.

• Visualise community marks on an OpenStreetMap-based map with filters.

• Provide SOS helplines and location sharing for emergency assistance.

• Offer an admin view of OSM metrics alongside Generated (demo) and Real (live) community signals.

• Defer finished EDA, unsupervised hotspot clustering, and predictive policing claims to later phases.

2. LITERATURE SURVEY

2.1 Background Study

GIS-based crime mapping literature (for example Chainey and Ratcliffe) establishes methods for visualising concentration of events and identifying hotspots. Environmental criminology links guardianship, lighting, and land use to both opportunity and fear. OpenStreetMap has become a standard volunteered geographic source for urban analytics when official spatial layers are incomplete or delayed. Crowdsourced safety audits, popularised by tools such as Safetipin, show that citizen ratings can complement police statistics by capturing perceived safety.

2.2 Existing System

• Personal safety apps (for example bSafe-style products) emphasise SOS alerts, trusted contacts, and audio/video evidence rather than city-scale street scoring.

• Standalone GIS notebooks or civic dashboards may score environments offline but lack a live anonymous reporting channel for women in the field.

• Official complaint portals record incidents after the fact and rarely expose interactive community maps for everyday route decisions.

2.3 Limitations of Existing System

• Emergency features without a accumulating spatial dataset leave little memory for others planning the same street later.

• Offline OSM analyses without a product front-end are inaccessible to non-technical citizens.

• Systems that require strong identity for every report raise privacy concerns for harassment-related disclosures.

• Closed proprietary safety scores are difficult to audit or reproduce in academic PBL settings.

2.4 Proposed System

The proposed SafeHer system addresses these limitations by combining: (i) a reproducible OSM multi-factor street score for Bengaluru BBMP boxes; (ii) anonymous feeling and incident reports with rate limiting; (iii) mobile and web clients sharing one backend; (iv) SOS

helplines and location share; and (v) an admin portal that separates Generated demonstration data from Real submissions so analysts do not confuse the two.



Fig. 2.1: Existing systems versus proposed SafeHer approach

2.5 Advantages of Proposed System

• Transparent nine-factor scoring that can be explained to faculty and civic partners.

• Dual client (mobile + web) lowers the barrier for users who cannot install an APK.

• Anonymous inserts with approved-only public reads balance contribution and moderation policy.

• Admin Generated-versus-Real labelling supports honest evaluation during demos.

• SOS and map features give immediate user value while the long-term dataset grows.

2.6 Feasibility Study

2.6.1 Technical Feasibility

The stack relies on mature open tools: Python with Overpass and pandas for dataset generation; Expo and React Native for Android builds; Next.js for the web portal; and Supabase (PostgreSQL + Row Level Security) for storage. All components run on standard student hardware and free-tier cloud resources, confirming technical feasibility for a PBL prototype.

2.6.2 Operational Feasibility

Citizens need only location permission and a few taps to contribute. Administrators use a password session to view dashboards. No hospital or police system integration is required for Phase 1, keeping operations within the student team's control.

2.6.3 Economic Feasibility

OSM data, Expo tooling, Next.js, and Supabase free tiers avoid licensed datasets and paid map SDKs for the prototype stage. Development cost is therefore limited to time and commodity devices, which is economically feasible for academic work.

2.7 Tools and Technologies Used

• Programming: Python 3.x, TypeScript / JavaScript

• Mobile: Expo (~54), Expo Router, React Native, AsyncStorage

• Web: Next.js 16 (App Router), React 19, Tailwind CSS, Leaflet / react-leaflet, Recharts

• Backend: Supabase (PostgreSQL, RLS, RPC), anonymous client key

• Data: Overpass API, pandas, Excel/JSON street exports

• Maps: OpenStreetMap tiles in WebView (mobile) and Leaflet (web)

2.8 Hardware and Software Requirements

Table 2.1: Hardware and Software Requirements

Category	Requirement
Processor	Intel i5 / AMD Ryzen 5 or higher (development)
RAM	8 GB minimum (16 GB recommended for dataset builds)
Storage	5 GB free for Node modules, Android SDK, and dataset files
Operating System	Windows 10/11 (team development); Android 10+ for field tests
Software	Node.js 20+, Python 3.10+, Expo CLI / EAS, Android Studio SDK
Cloud	Supabase project; optional Vercel hosting for web
Libraries	pandas, requests, @supabase/supabase-js, Leaflet, Recharts

3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Users / User Roles

• Citizen / woman traveller — reports feelings or incidents, views the community map, uses SOS.

• Admin / project researcher — authenticates to the admin portal for OSM KPIs and community analytics.

• Developer / student team — maintains dataset scripts, schema, and client applications.

3.1.1 Objectives

1. Build an OSM extraction and scoring pipeline for Bengaluru BBMP bounding boxes.

2. Publish a reusable scored street dataset (Excel and JSON) with Safe / Moderate / Unsafe labels.

3. Implement SafeHer mobile tabs for Home, Report, Map, and SOS.

4. Implement SafeHer web routes for reporting, map, emergency, and community summary.

5. Persist anonymous reports in Supabase with area labels and basic rate limiting.

6. Provide an admin portal separating Generated demo data from Real live reports.

7. Document Phase-1 limits honestly: EDA and hotspot clustering are out of scope for completion claims.

Scope: Phase 1 is a single-team prototype for data collection and platform demonstration. It does not include production citizen authentication, hospital information-system integration, finished unsupervised hotspot products, or city-wide operational deployment.

3.2 Assumptions and Dependencies

3.2.1 Assumptions

• Users grant location permission when reporting or using SOS.

• OSM coverage in Bengaluru is sufficient for a meaningful infrastructure baseline.

• Anonymous reporting with device-level rate limits is acceptable for the prototype ethics posture.

• Admin password in environment configuration is adequate for academic demonstration (not enterprise IAM).

3.2.2 Dependencies

• Overpass API availability for dataset regeneration.

• Supabase project reachability and anon key configuration in mobile/web clients.

• OpenStreetMap tile service for map rendering.

• Android device or emulator for Expo field tests; modern browser for web tests.

3.3 Functional Requirements

Table 3.1: Functional Requirements

ID	Requirement
FR1	The system shall fetch OSM highways and POIs per BBMP bounding box and compute street safety scores.
FR2	The system shall classify streets into Safe, Moderate, or Unsafe using documented thresholds.
FR3	The mobile app shall resolve GPS to a BBMP area name for display and storage.
FR4	The system shall accept anonymous safe / unsafe / incident reports with optional factors.
FR5	The system shall store reports in Supabase and expose approved reports to community maps.
FR6	The system shall rate-limit submissions per device to reduce spam (approximately five per hour).
FR7	The map shall filter pins by All / Safe / Unsafe / Incidents and show area context.
FR8	The SOS module shall list helplines and support calling plus location sharing.
FR9	The admin portal shall require a password session before dashboard access.
FR10	The admin analytics view shall distinguish Generated (synthetic) from Real reports.
FR11	The web site shall allow reporting and map use without citizen login.
FR12	The community summary shall aggregate counts useful for Home and Community screens.

3.4 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Requirement
NFR1	Usability: report flow completable in under one minute for the minimal path.
NFR2	Privacy: no citizen account; coordinates used for area naming, not public tracking profiles.
NFR3	Security: RLS enforces insert rules and approved-only public reads; admin via HttpOnly-style session cookie.
NFR4	Reliability: clients degrade gracefully when GPS or network is unavailable (refresh / retry messaging).
NFR5	Maintainability: shared BBMP area definitions between Python pipeline and TypeScript clients.
NFR6	Performance: map should remain interactive for hundreds of community pins on mid-range phones.
NFR7	Transparency: scoring weights and methodology visible in admin Methodology pages.
NFR8	Portability: Android APK via Expo; web via modern Chromium/Firefox/Safari browsers.

4. SYSTEM DESIGN

4.1 System Architecture

SafeHer follows a layered client–server design with an offline data-science component. Citizens use mobile or web clients. Both talk to Supabase for report writes and community reads. Static OSM-derived JSON under the web public data folder drives admin OSM dashboards. The Python Overpass pipeline regenerates the Excel/JSON baseline when needed.

Major building blocks:

1. OSM Dataset Pipeline — Overpass queries, nine-factor scoring, Excel/JSON export.

2. Supabase Backend — reports table, area_safety view, community_summary RPC, RLS policies.

3. Mobile Application — Expo tabs for Home, Report, Map (WebView Leaflet), SOS.

4. Web User Portal — landing, report form, map, emergency, community dashboard.

5. Web Admin Portal — password middleware, OSM KPIs, analytics, search, areas, methodology.

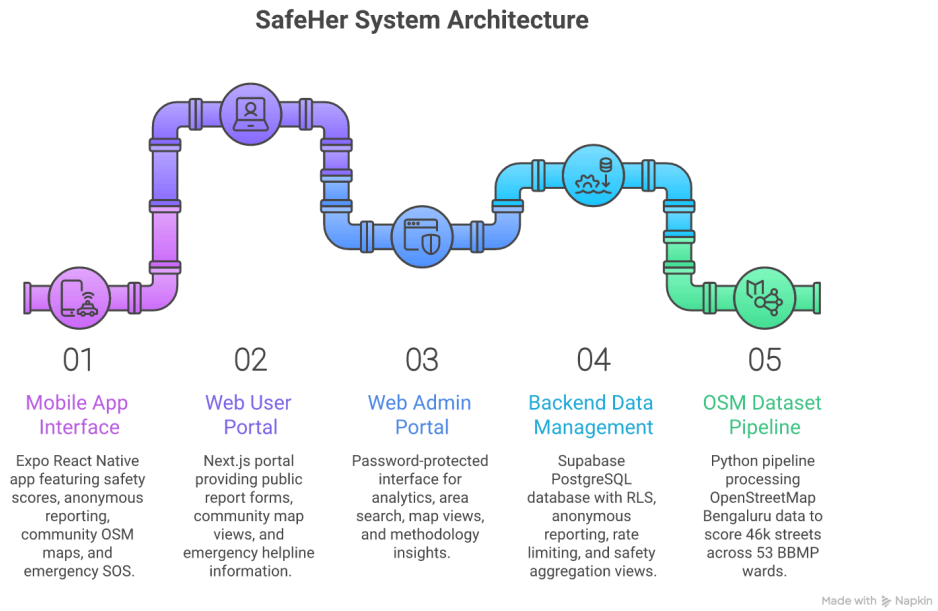


Fig. 4.1: High-level system architecture of SafeHer

4.2 System Perspective

From a product perspective, SafeHer is a civic sensing tool: infrastructure scores supply a baseline where crowdsourced volume is still low, while live reports gradually enrich place-level knowledge. The admin perspective treats Generated demo pins as scaffolding for empty maps and Real pins as the measurement of field adoption. Analytical ML modules are intentionally outside the Phase-1 system boundary so that architecture remains honest about completed scope.

4.3 Context Diagram

External actors are the citizen, the administrator, OpenStreetMap/Overpass, Supabase cloud, and telephony/SMS/WhatsApp channels used during SOS. The SafeHer system sits at the centre, mediating map tiles, report persistence, and emergency deep links without requiring a citizen login.

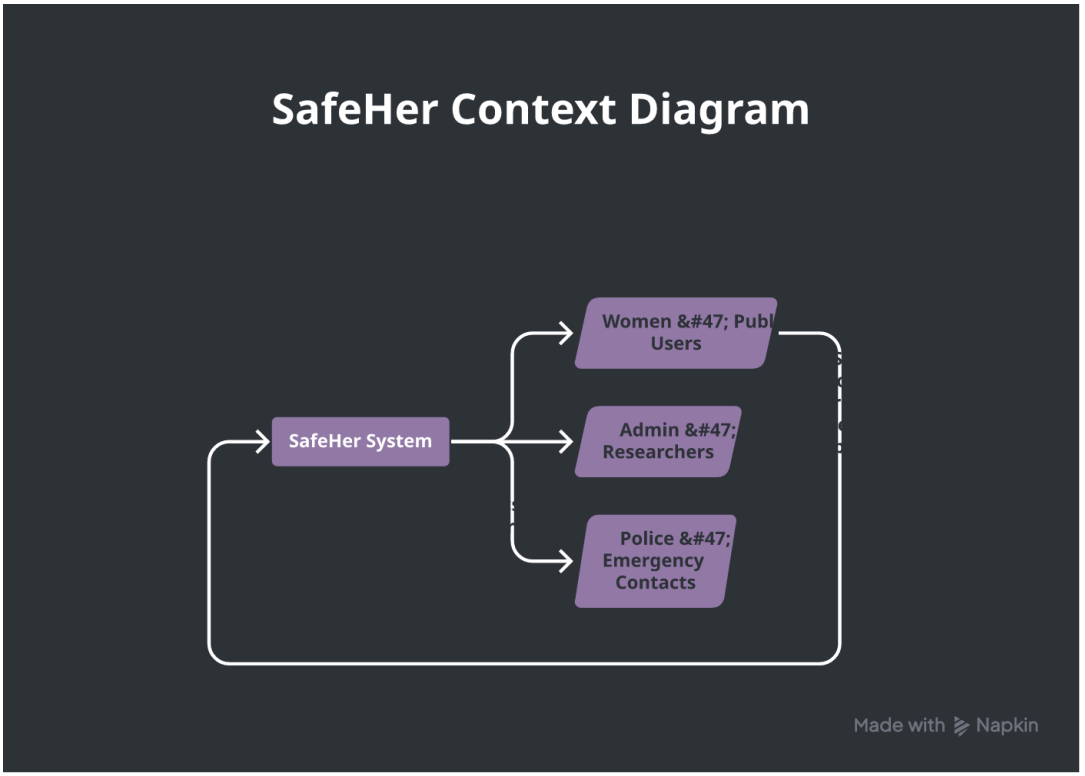


Fig. 4.2: System context diagram

5. DETAILED DESIGN

5.1 Data Flow Diagram – Level 0

At Level 0, citizens supply location and report choices; SafeHer persists approved community data and returns map/summary views. Administrators supply credentials and receive analytics. The offline pipeline consumes OSM and produces scored streets.

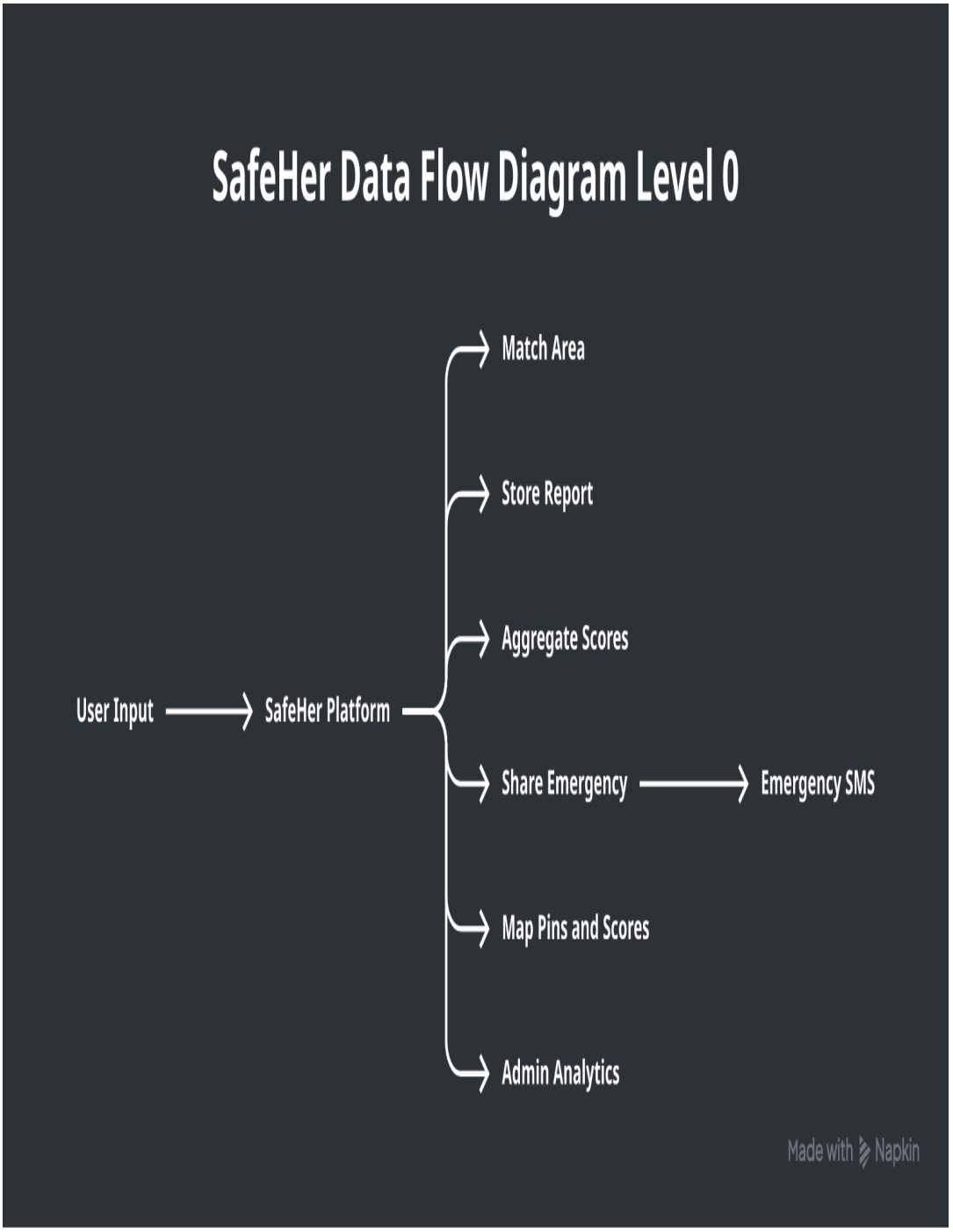


Fig. 5.1: Data flow diagram – Level 0

5.2 Data Flow Diagram – Level 1

Level 1 expands internal processes: area matching, report validation and insert, community aggregation, OSM scoring, and admin session checks. Data stores include the reports table, static street JSON, and local trusted-contact preferences on device.

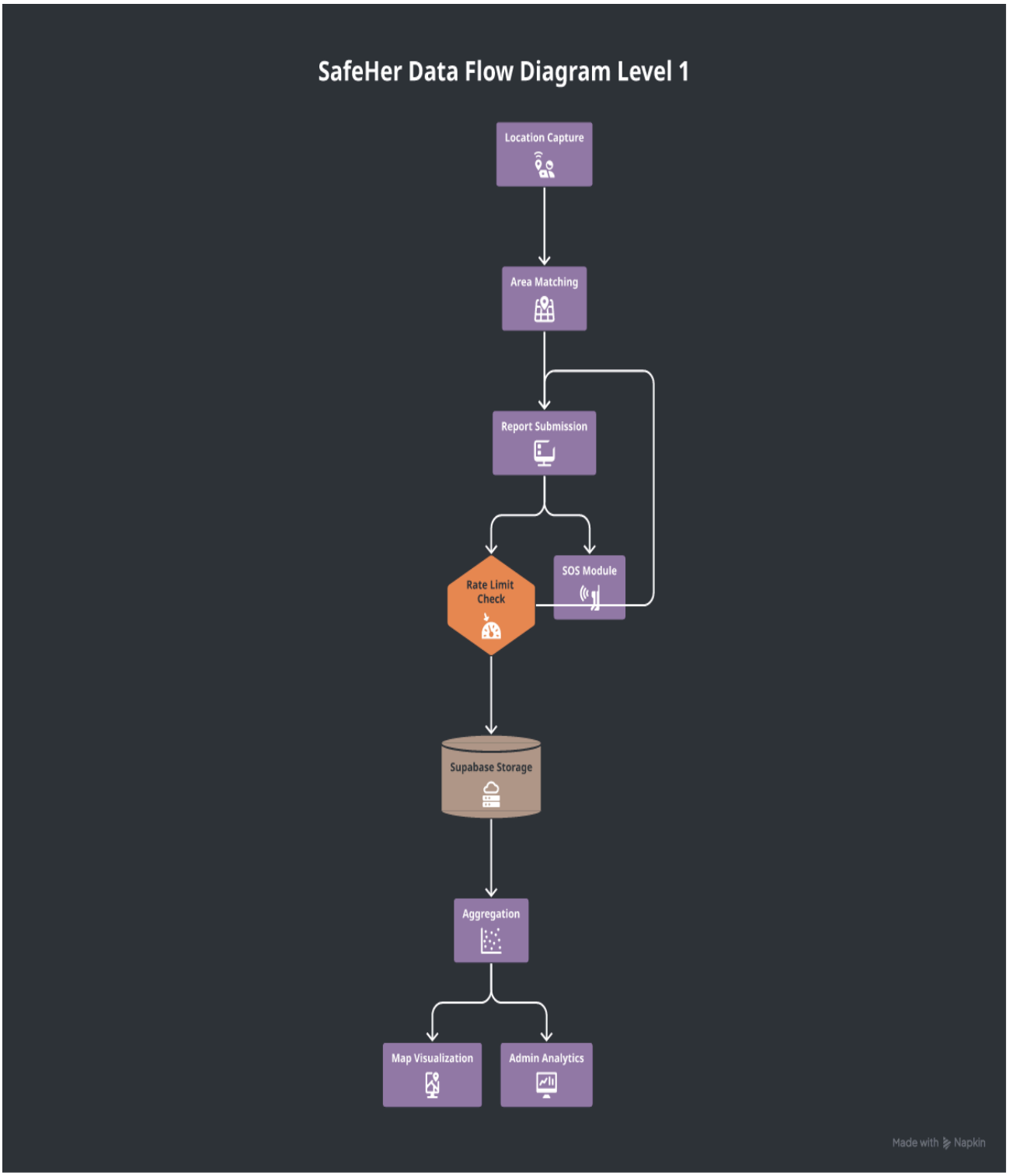


Fig. 5.2: Data flow diagram – Level 1

5.3 Use Case Diagram

Primary use cases for the citizen are Confirm Location, Submit Feeling Report, Submit Incident Report, Browse Safety Map, Call Helpline, and Share Live Location. Administrator use cases include Login, View OSM Dashboard, Inspect Community Analytics, Search Streets/Areas, and Read Methodology.

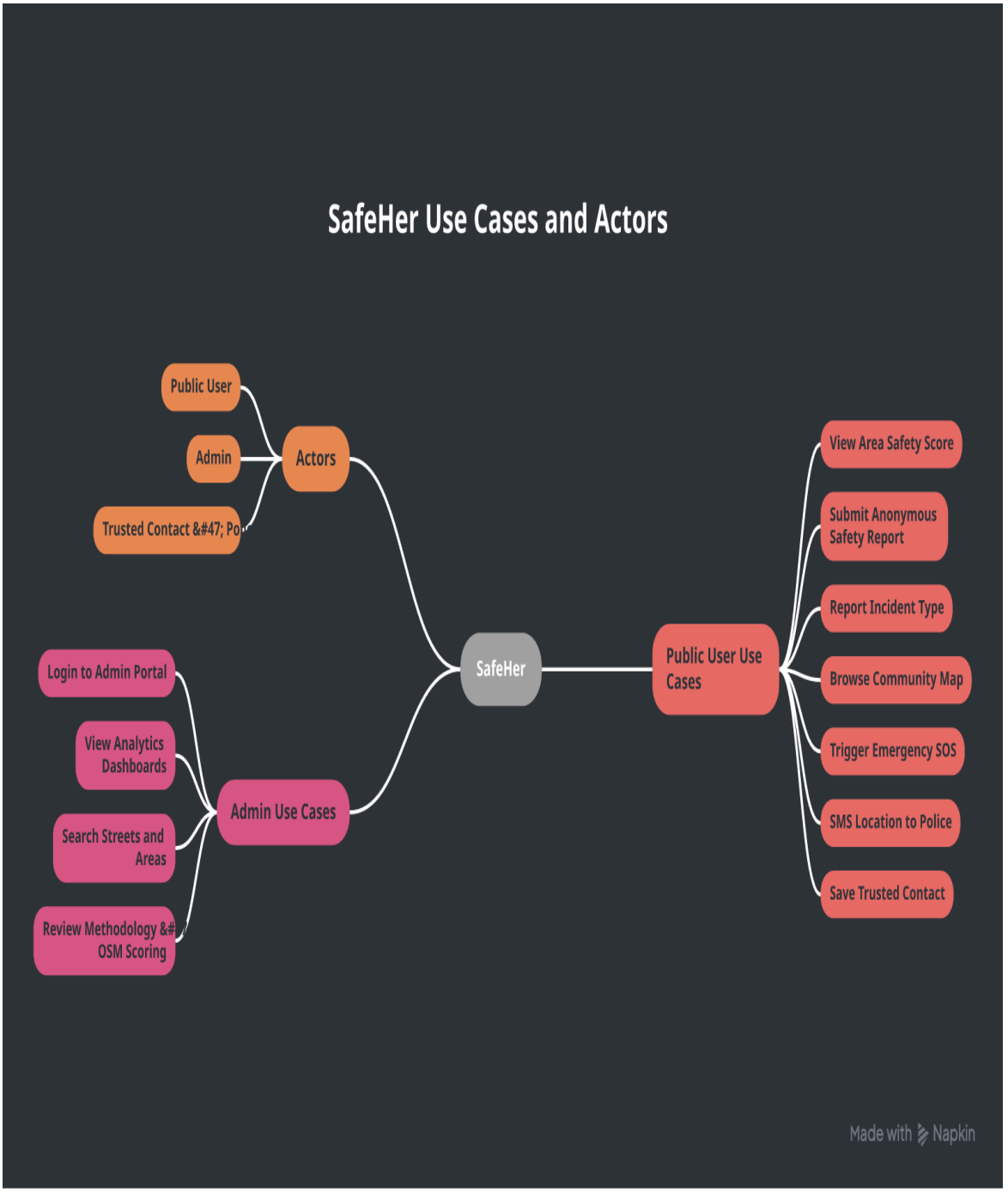


Fig. 5.3: Use case diagram

5.4 Activity Diagram – Emergency SOS

The SOS activity begins when the user opens the Emergency tab, optionally refreshes GPS, chooses Call 112 and share, share-to-police SMS, or a specific helpline card. Live tracking can keep updating coordinates for repeated shares. Trusted contact numbers remain on-device storage only.

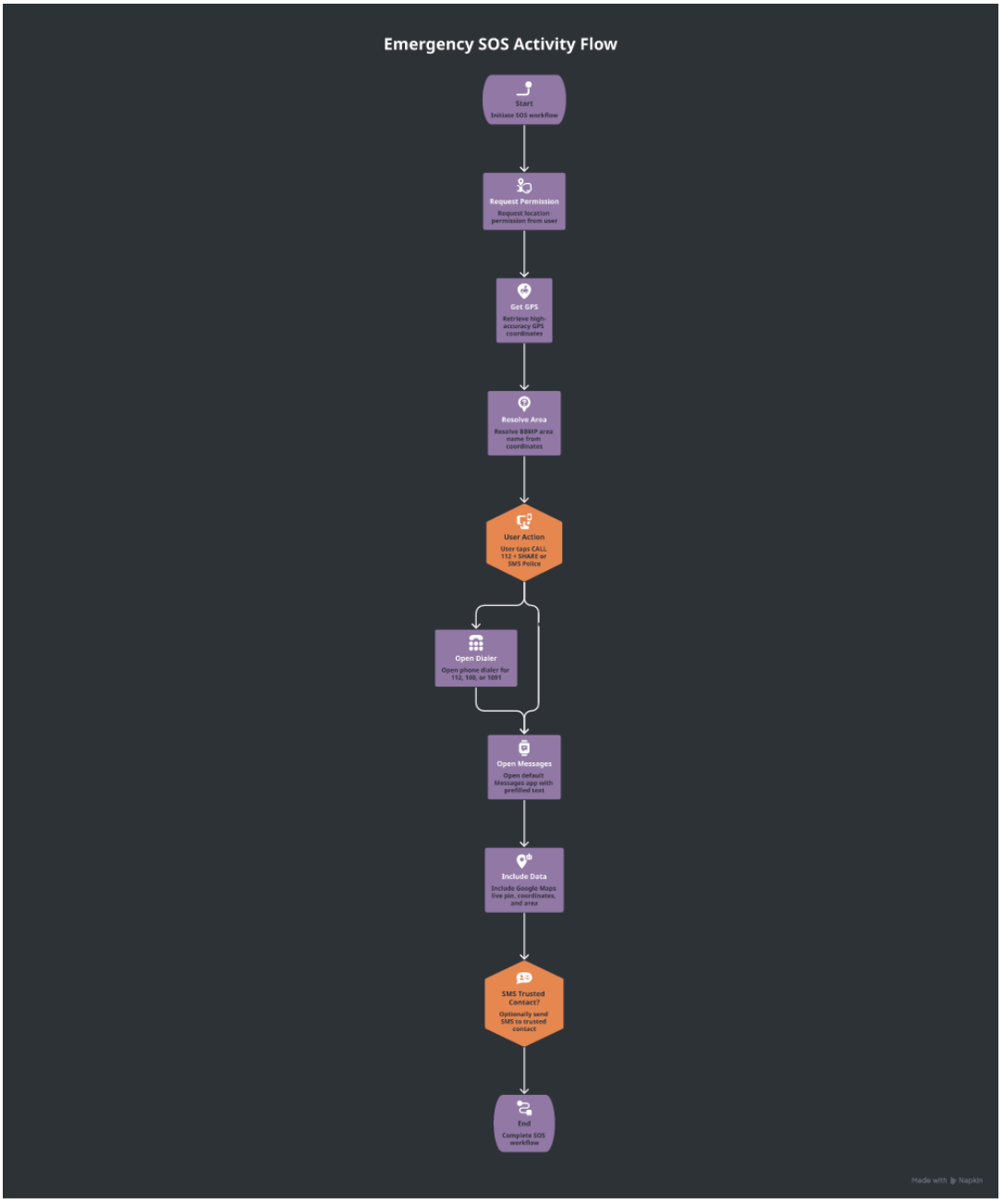


Fig. 5.4: Activity diagram – Emergency SOS flow

5.5 Sequence Diagram – Report Submission

The report sequence involves the mobile/web UI, area matcher, Supabase client, RLS policies, and optional community_summary refresh. After insert, the UI shows a thank-you state and offers navigation to the map or a new report.

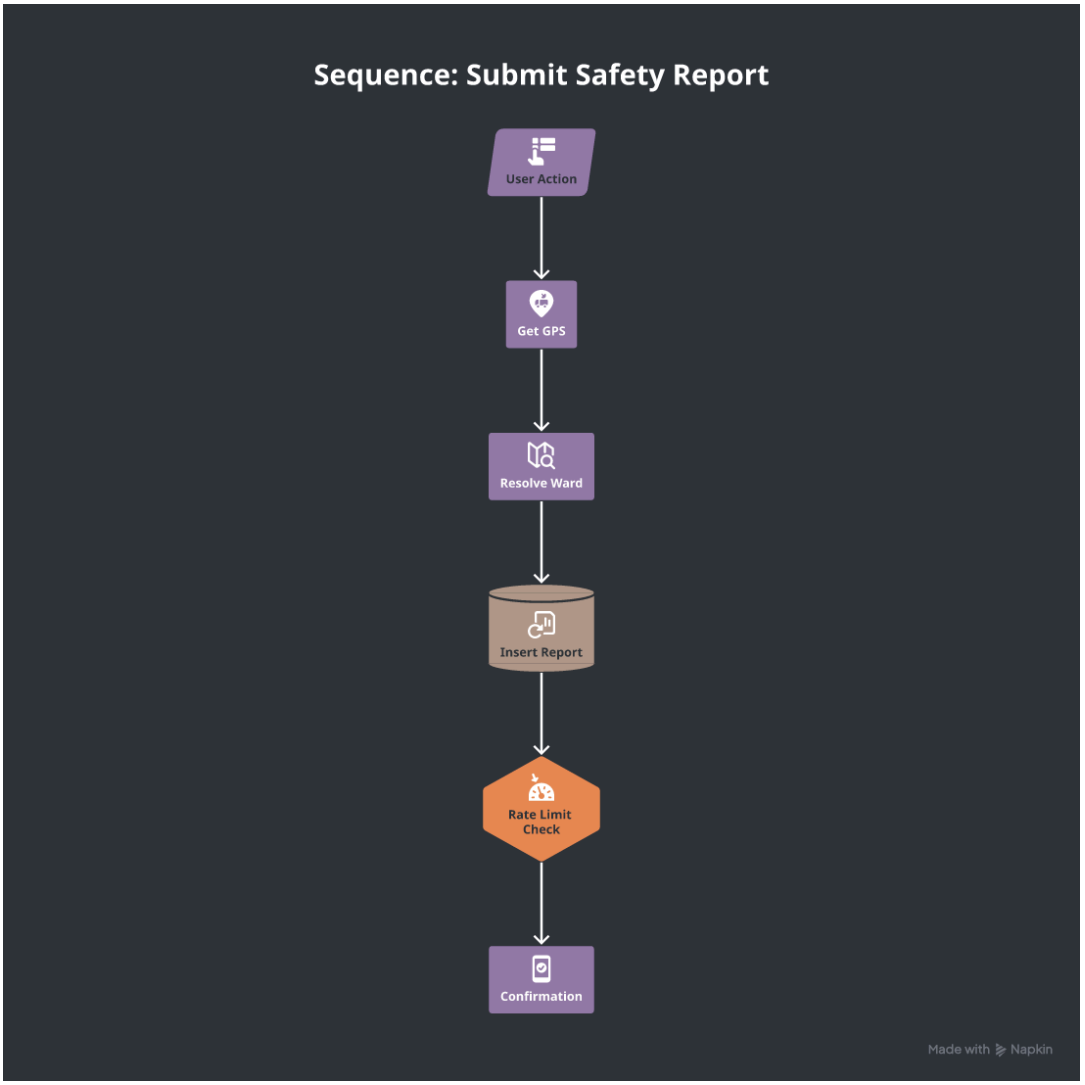


Fig. 5.5: Sequence diagram – anonymous report submission

6. IMPLEMENTATION

6.1 Implementation Methodology

Implementation followed an iterative full-stack approach organised into four workstreams: (1) OSM dataset pipeline and scoring validation; (2) Supabase schema, RLS, and RPC functions; (3) Expo mobile application; (4) Next.js user and admin portals. Shared BBMP bounding boxes in TypeScript/Python kept area names consistent across offline scoring and online GPS matching.

6.2 Dataset and Feature Description

Bengaluru is partitioned into 53 BBMP area boxes. For each box, the Python script queries Overpass for highway geometries and POIs (police, hospitals, bus stops, CCTV nodes, lamps, and related amenities). Distances and presence flags feed a weighted score. Thresholds map scores to Safe (≥ 7), Moderate (≥ 4), and Unsafe (< 4).

Table 6.1: OSM safety scoring factor weights

Factor	Weight
Police proximity	0.20
Hospital proximity	0.15
Bus stop proximity	0.15
Lighting indicators	0.15
CCTV presence	0.10
Crowd / activity proxies	0.10
Commercial activity	0.05
Night amenity indicators	0.05
Road class	0.05

Table 6.2: Bengaluru OSM dataset summary metrics

Metric	Value
Total streets scored	46,424
BBMP areas covered	53
Safe streets	7,855
Moderate streets	33,431
Unsafe streets	5,138
Average safety score	5.78
Dataset generation date	2026-05-24

6.3 Module-wise Implementation

6.3.1 Mobile Application

The Expo Router tabs are Home (area verdict and CTAs), Report (multi-step wizard), Map (Leaflet HTML inside WebView with filters), and SOS (helplines, trusted contact, call and share helpers). Reports post to Supabase using the anon client; device identifiers support rate limiting without creating user accounts.

6.3.2 Web User Portal

Public routes under the user layout expose landing, report, map, emergency, and community pages. The community dashboard merges synthetic demonstration records with live approved reports and labels the mix so evaluators understand demo scaffolding.

6.3.3 Web Admin Portal

Admin login validates ADMIN_PASSWORD and sets a session cookie consumed by Next.js middleware. Dashboards read static OSM JSON for street KPIs and query Supabase for community analytics, search, and area views. A methodology page documents scoring.

6.3.4 Supabase Backend

The reports table stores latitude, longitude, area name, is_safe, rating, factor booleans, comment, time_of_day, device_id, status, report_kind (feeling or incident), and incident_type. RLS allows anonymous insert and public select of approved rows. RPCs provide community_summary and recent-device report counts.

6.4 Key Code Snippets

Illustrative weighted scoring (Python pipeline):

```
weights = {
    "police": 0.20, "hospital": 0.15, "bus": 0.15,
    "lighting": 0.15, "cctv": 0.10, "crowd": 0.10,
    "commercial": 0.05, "night": 0.05, "road": 0.05,
}
score = sum(component(row, k) * w for k, w in weights.items())
level = "Safe" if score >= 7 else "Moderate" if score >= 4 else "Unsafe"
```

Illustrative report insert contract (client):

```
await supabase.from("reports").insert({
  lat, lon, area_name, is_safe, rating,
  report_kind, incident_type, device_id, status: "approved",
});
```

6.5 Visuals

The following figures are field captures of the SafeHer Android build exercising Home, Report, Map, and Emergency SOS flows.

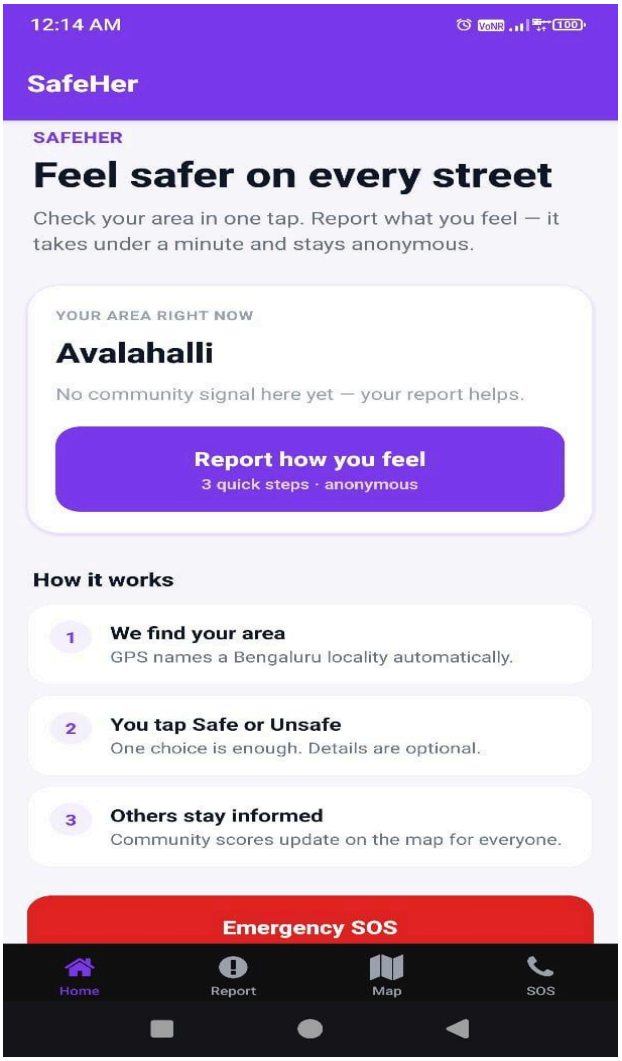


Fig. 6.1: SafeHer Home – area status and call to report

SafeHer home screen. Headline “Feel safer on every street,” current area Avalahalli, and a purple Report how you feel button for a short anonymous report. Emergency SOS is visible lower down; tabs are Home, Report, Map, SOS.

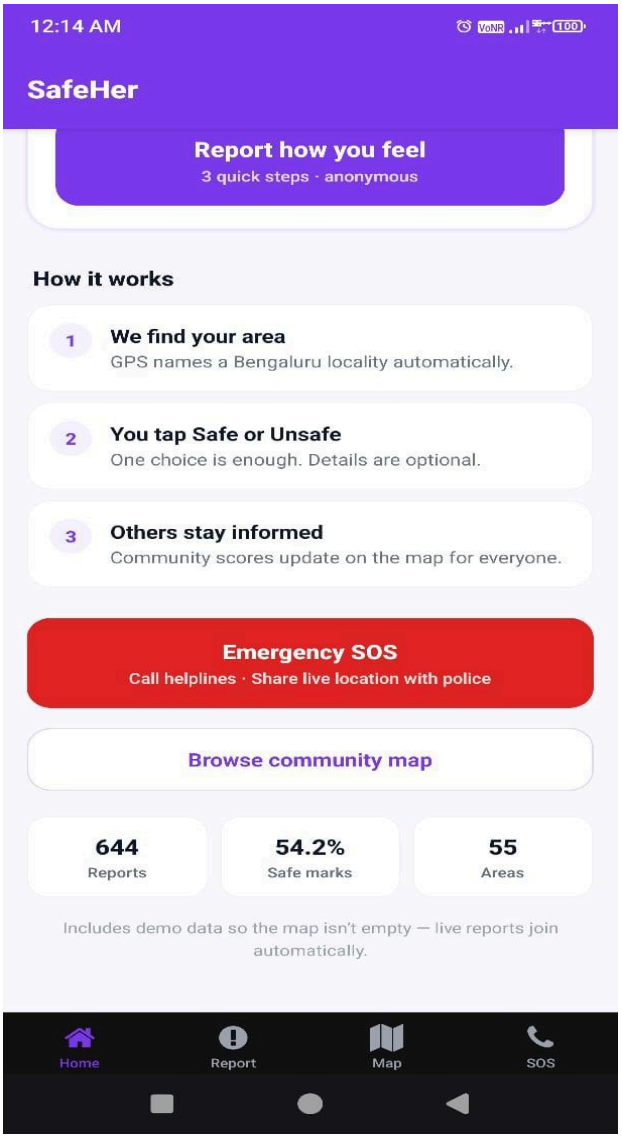


Fig. 6.2: SafeHer Home – how it works and community summary

Same home screen after scroll: three "How it works" steps, SOS with helpline/location share, link to the community map, and summary stats (reports / safe marks / areas).

Note that demo data is included so the map isn't empty.

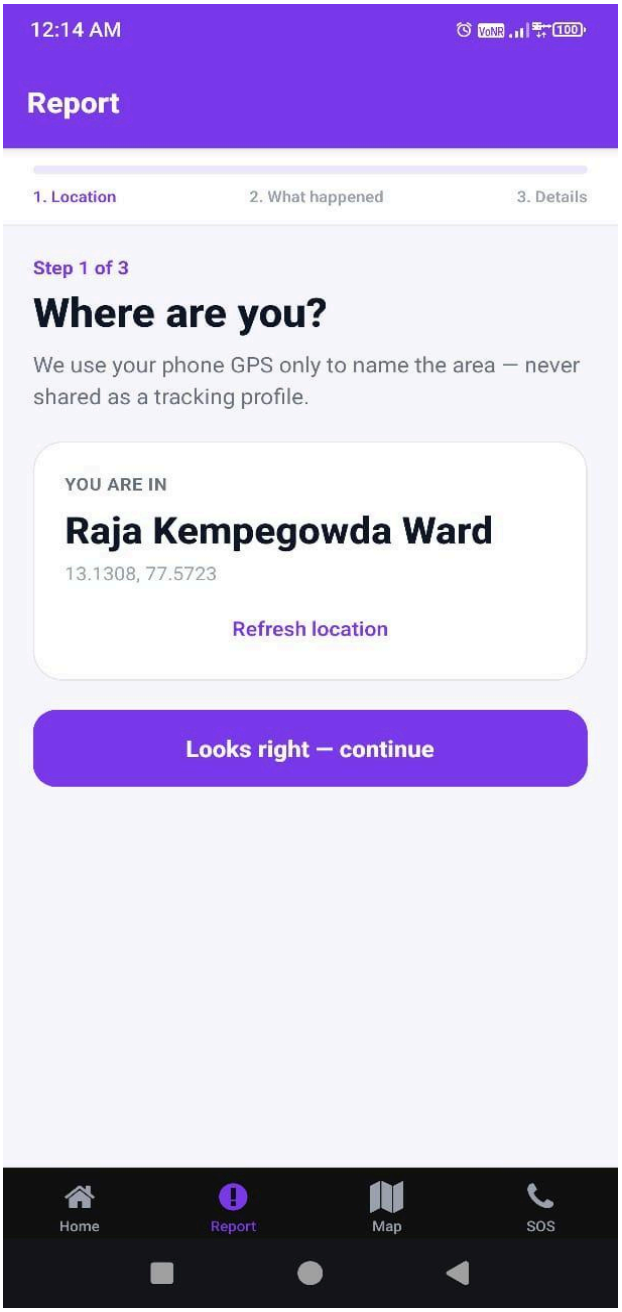


Fig. 6.3: Report wizard – Step 1 location confirmation

First step of the report wizard. GPS names Raja Kempegowda Ward (with coordinates). User can refresh location or tap Looks right — continue. Text says GPS is only used to name the area, not for tracking.

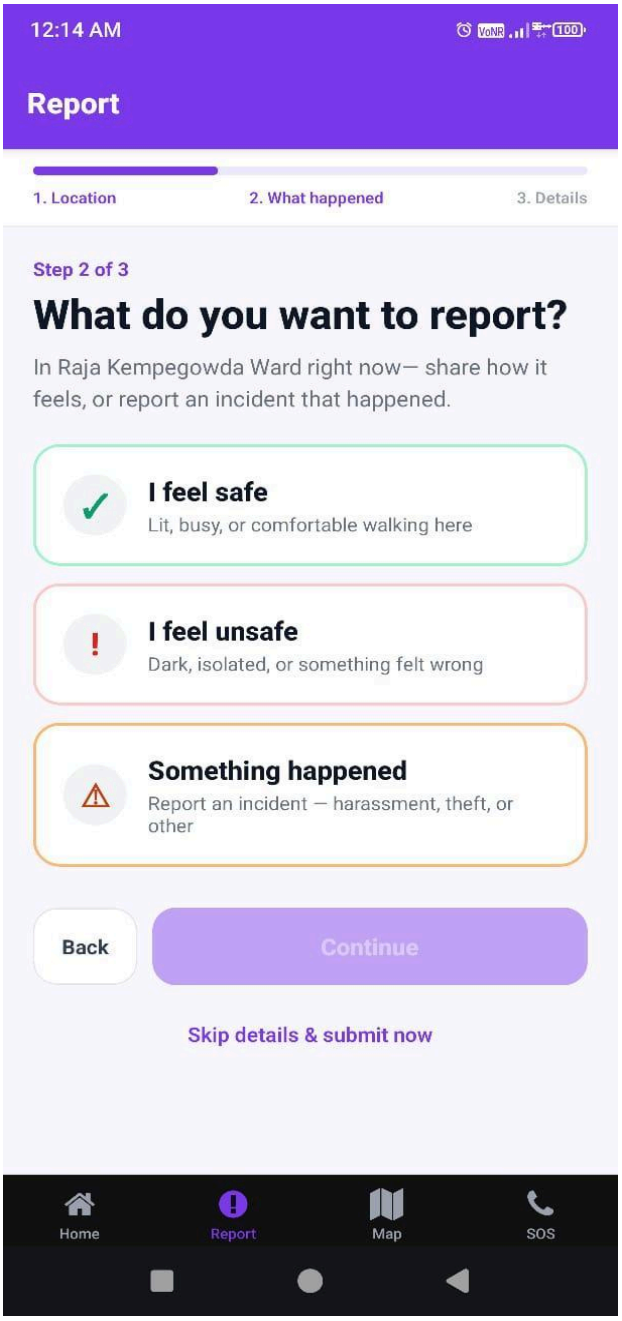


Fig. 6.4: Report wizard – Step 2 feeling or incident choice

User picks I feel safe, I feel unsafe, or Something happened. Each choice has a short hint. Back / Continue, plus Skip details & submit now.

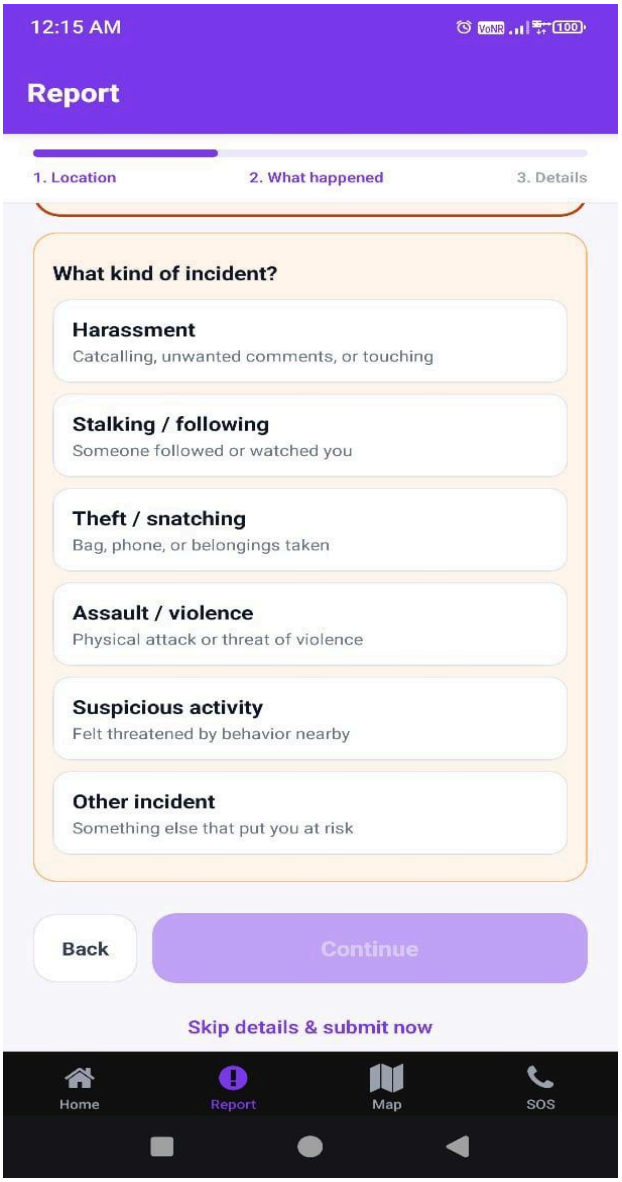


Fig. 6.5: Report wizard – incident type selection

User picks I feel safe, I feel unsafe, or Something happened. Each choice has a short hint. Back / Continue, plus Skip details & submit now.

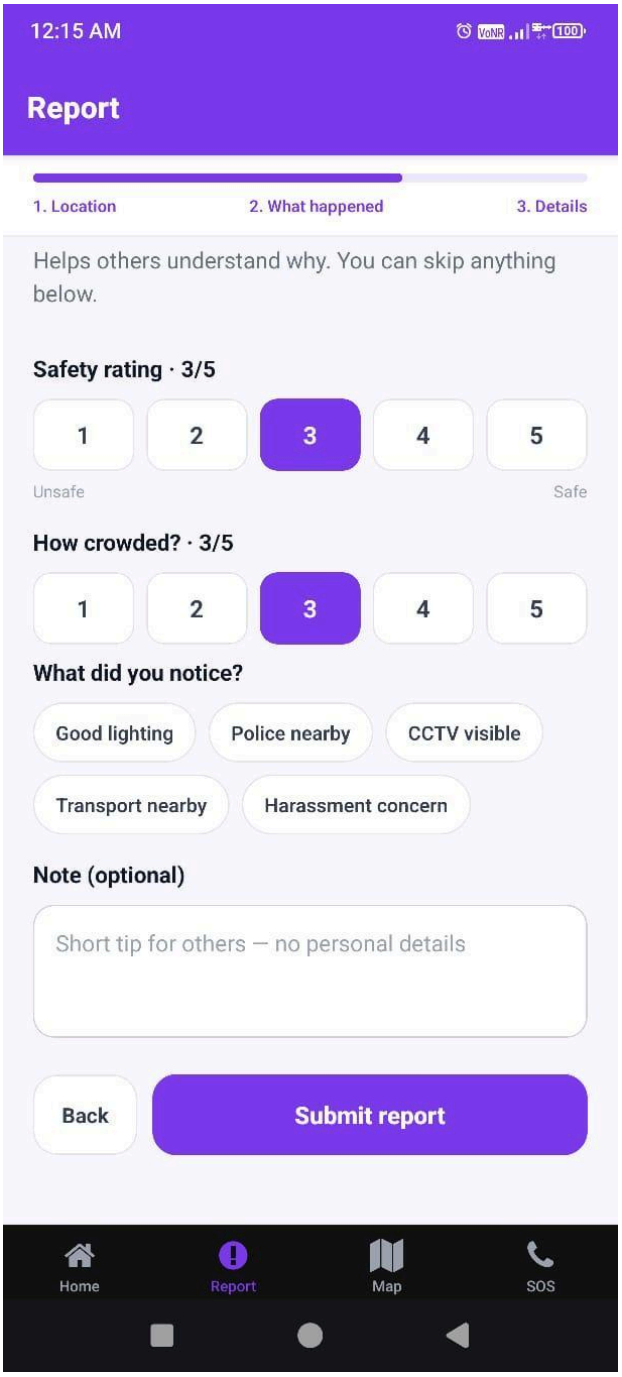


Fig. 6.6: Report wizard – optional details and factors

Optional details: 1–5 safety rating, crowd level, tags (lighting, police, CCTV, transport, harassment concern), and a short note. Fields can be skipped before Submit report.

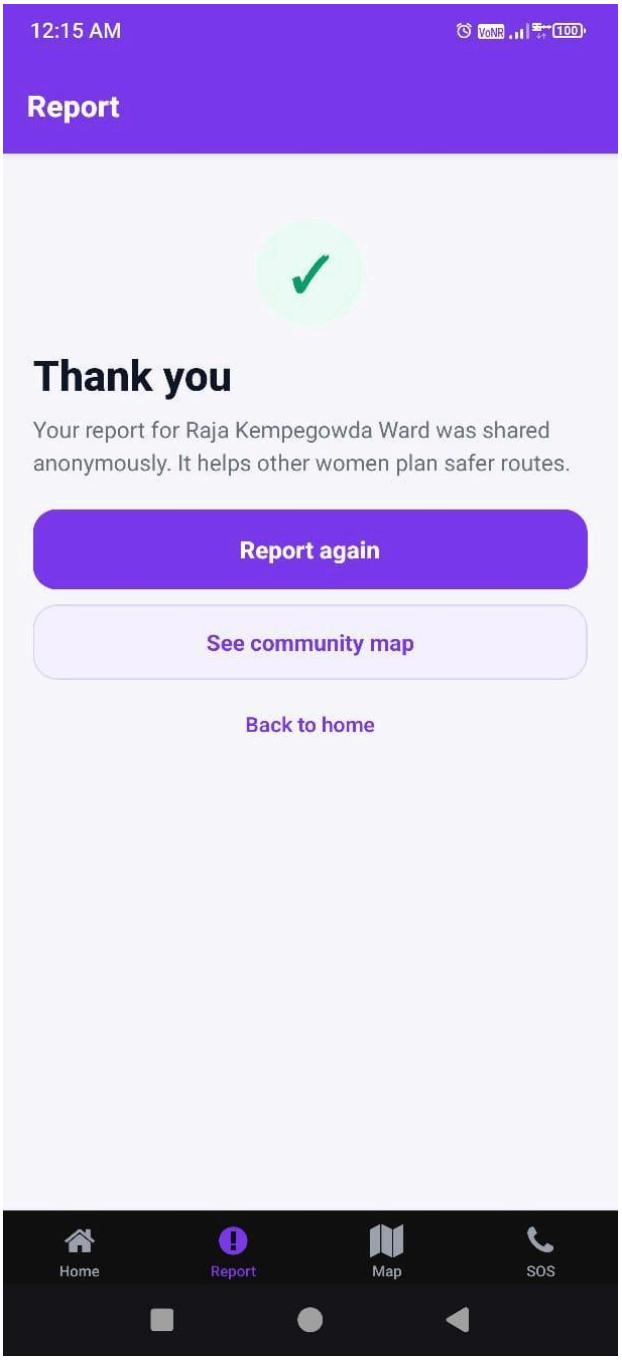


Fig. 6.7: Report submission success screen

Thank-you screen after save. Confirms the report for the ward was shared anonymously, then offers Report again, See community map, or Back to home.

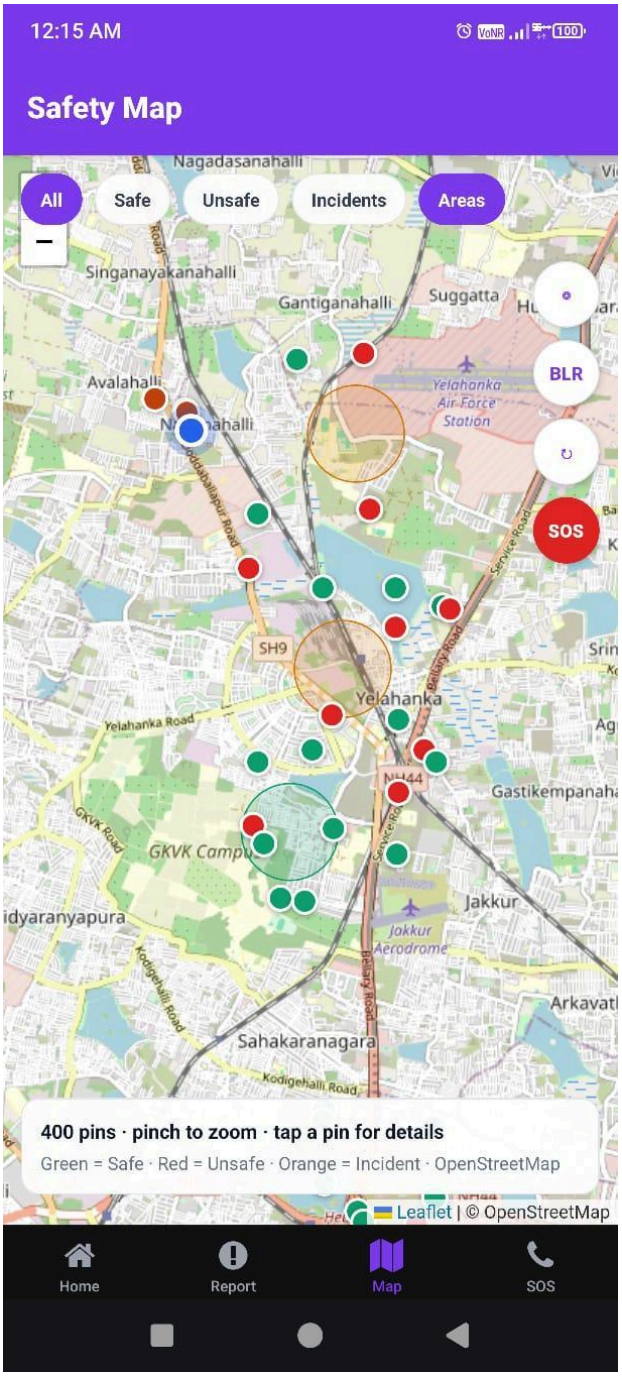


Fig. 6.8: Safety Map with filters and community pins

Thank-you screen after save. Confirms the report for the ward was shared anonymously, then offers Report again, See community map, or Back to home.

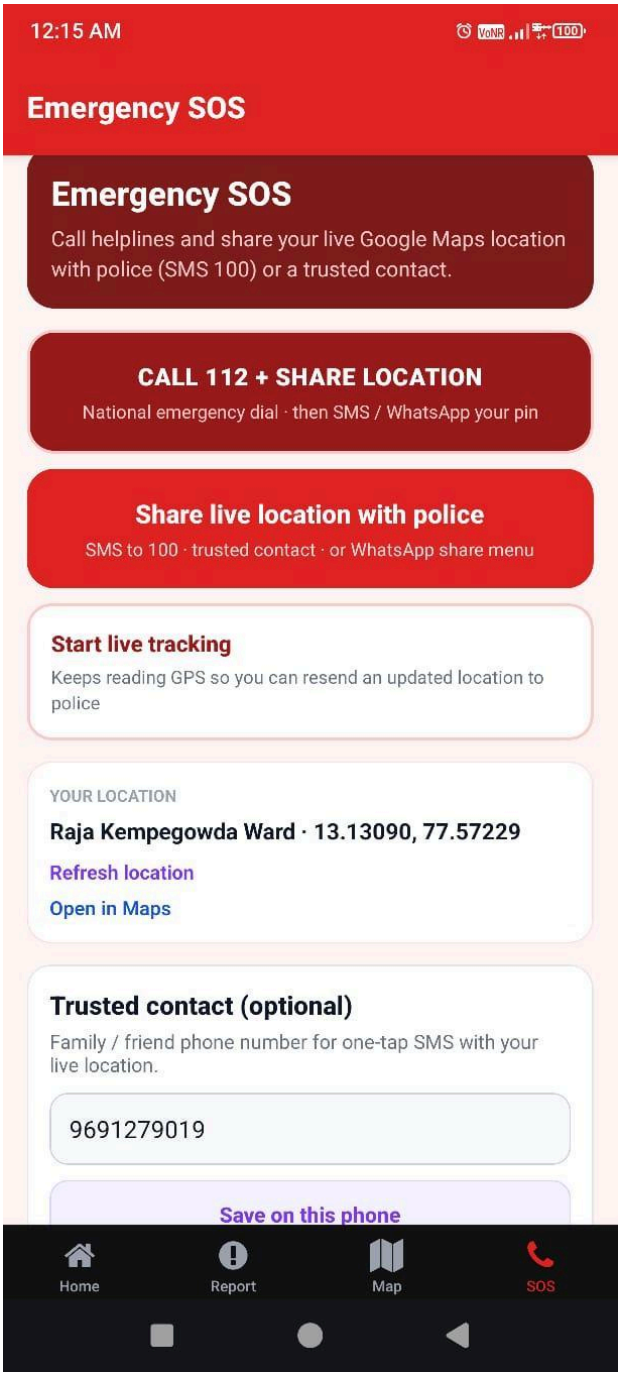


Fig. 6.9: Emergency SOS – call and live location share

SOS hub: Call 112 + share location, share live location with police (SMS 100), live tracking, current ward/coordinates, and optional trusted contact saved on the phone.

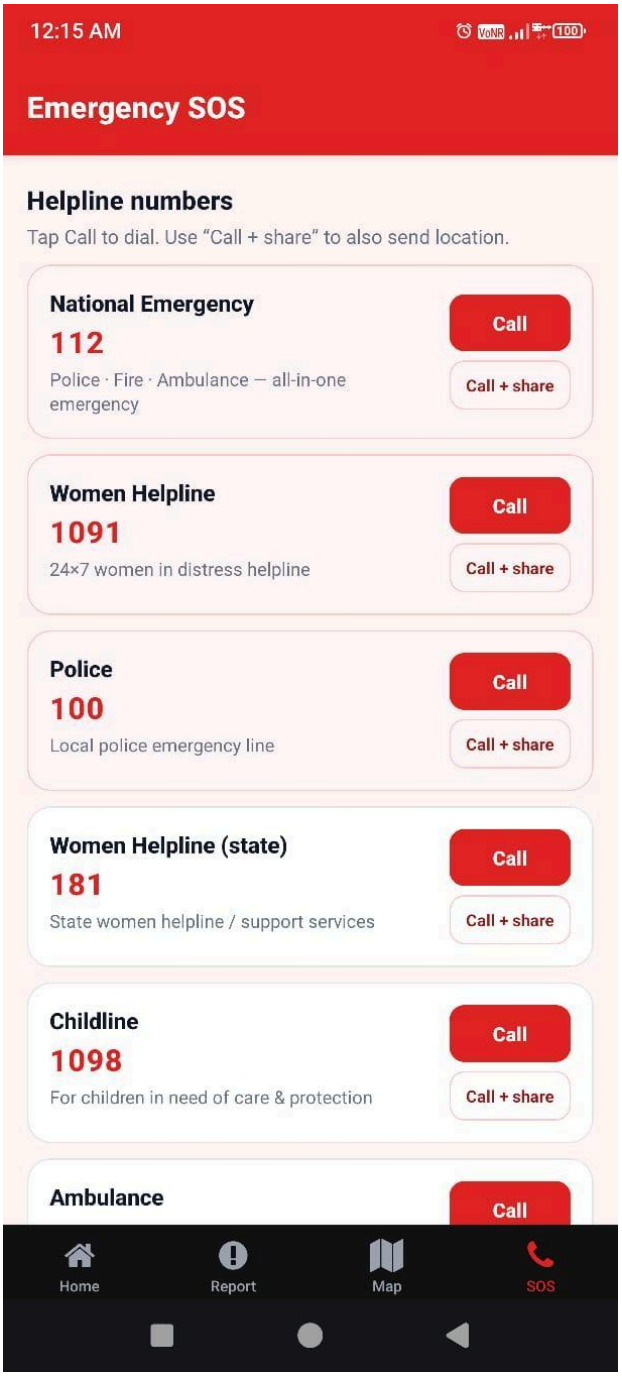


Fig. 6.10: Emergency SOS – helpline directory

Helpline cards (112, 1091, 100, 181, 1098, ambulance) with Call or Call + share so location can be sent with the call.

7. SOFTWARE TESTING

7.1 Testing Methodology

Testing combined manual functional checks on Android devices, browser smoke tests for the web portal, SQL verification of RLS behaviour, and scripted validation of scoring consistency between Python and TypeScript helpers. Emphasis was on critical user journeys: report submission, map filters, SOS dial/share, and admin auth gates.

7.2 Unit Testing

Unit-level checks covered BBMP point-in-box area matching, safety level thresholds, and scoring weight totals. Dataset conversion scripts were validated by comparing summary counts in Excel against generated JSON summary files used by the admin UI.

7.3 Test Cases and Results

Table 7.1: Test Cases and Results

TC ID	Test Case	Expected Result	Status
TC01	GPS resolves to a BBMP area name on Home	Named area shown (or clear empty state)	Pass
TC02	Submit minimal safe report without details	Thank-you screen; row in Supabase	Pass
TC03	Submit incident with type Harassment	Stored with report_kind=incident	Pass
TC04	Map filter Unsafe hides safe pins	Only unsafe/incident-relevant pins remain	Pass
TC05	SOS Call 112 opens dialer	tel: intent launched	Pass
TC06	Share location composes SMS/Maps link	Share sheet or SMS with coordinates	Pass
TC07	Admin without cookie blocked from /admin/dashboard	Redirect to login	Pass
TC08	Admin with valid password reaches dashboard	OSM KPIs visible	Pass
TC09	Rate limit after repeated inserts	Further inserts rejected by RPC/policy	Pass
TC10	Public select cannot read pending/rejected rows	Only approved visible on map APIs	Pass

8. RESULTS AND DISCUSSION

8.1 Deliverables Achieved

Phase 1 produced a complete data-collection stack: a scored OSM street corpus for Bengaluru BBMP areas; working SafeHer mobile and web clients; Supabase persistence with privacy-minded RLS; and an admin portal for monitoring. Field screenshots in Chapter 6 confirm that the report wizard, map filters, and SOS helplines operate on device as designed.

8.2 Dataset Outcomes

Of 46,424 scored streets, most fall in the Moderate band, which is expected for a multi-factor model that rewards proximity to guardianship amenities without claiming crime prediction. The Unsafe minority highlights corridors where infrastructure proxies are weak and where future crowdsourced volume will be especially valuable.

8.3 Platform Behaviour

Anonymous reporting with optional detail fields keeps interaction cost low. Merging synthetic community pins for demonstration prevents empty-map demos while UI copy discloses that live reports join automatically. This honesty is important for viva evaluation and for any civic partner reviewing the prototype.

8.4 Discussion and Limitations

Absolute hotspot claims would be premature: Real report volume is still growing, OSM proxies are imperfect substitutes for lived experience, and unsupervised clustering has not been executed as a finished analytical product. Admin authentication via shared password suits a classroom prototype but must be replaced before any public-sector deployment. Despite these limits, Phase 1 meets its stated goal—establishing a dual data-collection architecture that later EDA and hotspot modules can consume.

9. CONCLUSION

This PBL project successfully designed and implemented SafeHer as a Phase-1 system for women's street safety data collection in Bengaluru. By pairing an auditable OSM street safety baseline with anonymous mobile and web reporting, community maps, SOS assistance, and an admin monitoring portal, the team delivered a coherent platform that is usable today and extensible for future analytics.

The work demonstrates that open geographic data and privacy-conscious crowdsourcing can be combined within a student timeline without overstating analytical maturity. Subsequent phases can layer EDA, clustering-based hotspot identification, and richer moderation workflows onto the foundations documented in this report.

10. FUTURE ENHANCEMENTS

1. Complete exploratory data analysis on Real reports joined with OSM street attributes.

2. Apply unsupervised methods (for example K-Means or DBSCAN) for hotspot candidate detection with careful validation.

3. Build moderated heatmaps and time-of-day views once Real volume is statistically meaningful.

4. Replace shared admin password with role-based authentication and audit logs.

5. Add optional verified volunteer roles for trusted on-ground audits.

6. Improve offline caching of map tiles and queued report submission under poor network.

7. Conduct structured usability studies with women students and refine SOS flows accordingly.

8. Explore partnerships for validating proxies against anonymised official statistics where legally permitted.

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5. Next.js Documentation — https://nextjs.org/docs
6. Supabase Documentation — https://supabase.com/docs
7. Leaflet — https://leafletjs.com/
8. React Native Documentation — https://reactnative.dev/
9. United Nations Sustainable Development Goals — https://sdgs.un.org/goals
10. Safetipin — https://safetipin.com/ (crowdsourced safety audit reference)
11. BBMP / Bengaluru civic geographic references used for ward bounding-box design.

APPENDIX A: SUSTAINABLE DEVELOPMENT GOALS (SDG)

This project is closely aligned with the global initiative of the United Nations under Sustainable Development Goal 5 (SDG 5): Gender Equality, which focuses on achieving gender equality and empowering all women and girls.

One of the key aspects of SDG 5 is addressing violence against women in both public and private spaces. The project specifically relates to:

1. **Target 5.2 —** eliminate all forms of violence against women and girls, including harassment and unsafe conditions in public areas such as streets, transportation systems, and marketplaces.
2. **Indicator 5.2.2 —** measures the proportion of women and girls subjected to sexual violence by persons other than an intimate partner, based on location and context of occurrence.

APPENDIX B: PROJECT SYNOPSIS

B1. INTRODUCTION

Women's safety in public spaces has become a critical concern in modern society, particularly in urban areas where population density and mobility are high. Incidents of harassment and violence occurring in locations such as streets, public transportation, parks, and marketplaces not only threaten physical safety but also impact the psychological well-being and freedom of women. These challenges often limit their participation in education, employment, and social activities, thereby affecting overall societal development.

Research in crime analysis has shown that such incidents are not randomly distributed but tend to occur in specific locations and under certain conditions. Studies by Lawrence Sherman highlight that crimes are often concentrated in particular areas, known as hotspots. Similarly, the work of Spencer Chainey and Jerry Ratcliffe emphasizes the importance of spatial analysis and Geographic Information Systems (GIS) in identifying patterns in crime data and supporting preventive decision-making.

In recent years, several technological solutions have been developed to enhance women's safety. Applications such as bSafe provide emergency assistance through features like SOS alerts and real-time location sharing. Similarly, My Safetipin uses crowd-sourced data to evaluate the safety of different locations. While these applications are useful in emergency situations, they primarily focus on reactive measures and depend heavily on user interaction.

A key limitation of existing systems is the lack of a preventive approach. Most solutions do not analyze historical incident data to identify patterns or predict high-risk areas. With advancements in data analytics and spatial technologies, it is now possible to move beyond reactive systems and develop solutions that can proactively identify unsafe locations.

This project addresses this gap by proposing a data-driven system that analyzes incident data to detect hotspots and visualize unsafe areas. By applying clustering techniques and spatial analysis, the system identifies regions where incidents are more likely to occur. The results are presented through intuitive visualizations, enabling users to make safer decisions and assisting authorities in planning effective safety measures.

Furthermore, the project aligns with the global initiative of the United Nations under Sustainable Development Goal 5, which aims to eliminate violence against women and ensure safe and

inclusive public spaces. By focusing on prevention and data-driven insights, this work contributes to building safer urban environments and promoting gender equality.

B2. OBJECTIVES

The main objective of this project is to develop a data-driven system that analyzes incidents of women harassment in public spaces and identifies high-risk areas to support preventive safety measures.

The specific objectives of the project are as follows:

1. **To analyze incident data related to women harassment —** the project aims to collect and examine data such as geographic location, time of occurrence, and type of incident. By studying this data, the system can identify patterns and trends that help in understanding how and where such incidents occur.
2. **To identify high-risk areas (hotspots) —** one of the key objectives is to detect areas where incidents are frequently reported. These areas, known as hotspots, are identified using data analysis techniques, enabling focused attention on locations that require improved safety measures.
3. **To apply spatial analysis techniques —** the project utilizes concepts from Geographic Information Systems to study the geographic distribution of incidents. This helps in understanding how location plays a role in safety and risk.
4. **To visualize unsafe zones effectively —** the system aims to represent identified hotspots using maps and graphical tools. Visualization techniques such as Heatmaps help users easily interpret complex data and recognize unsafe areas.
5. **To implement clustering for hotspot detection —** the project uses techniques like K-Means Clustering to group similar incident locations and automatically identify areas with high concentration of incidents.
6. **To study patterns based on time and location —** the system analyzes temporal patterns (such as day/night trends) along with spatial data to understand when incidents are more likely to occur.
7. **To support preventive safety and urban planning —** the insights generated from the system can help authorities improve infrastructure such as lighting, surveillance, and patrolling, contributing to safer public spaces.

8. **To align with Sustainable Development Goal 5 (Gender Equality) —** the project aims to contribute towards reducing violence against women by providing data-driven insights that support safer environments.

B3. LITERATURE SURVEY

Several research studies have explored the use of technology and spatial analysis to improve public safety, especially for women.

One of the important works in crime analysis is by Spencer Chainey and Jerry Ratcliffe (2005), who explained how Geographic Information Systems (GIS) can be used for crime mapping and hotspot detection. Their work highlights that analyzing geographic patterns of incidents can help authorities identify high-risk areas and take preventive measures.

Similarly, Lawrence Sherman (1989) introduced the concept of crime hotspots, showing that crimes are often concentrated in specific locations rather than being randomly distributed. This idea forms the foundation for hotspot identification techniques used in modern safety systems.

In the context of urban safety, Hannah Nehme et al. (2017) studied how environmental and spatial factors influence women's safety in cities. Their research emphasizes that factors such as lighting, crowd density, and location characteristics play a major role in determining safety levels.

Modern applications such as My Safetipin use crowd-sourced data and spatial analysis to provide safety scores for different locations. Similarly, apps like bSafe focus on emergency response through SOS alerts and real-time tracking.

However, research indicates that most existing systems are reactive and do not fully utilize historical incident data for predictive analysis. There is a need for systems that combine spatial analysis, data clustering, and visualization to identify unsafe areas in advance.

This project builds upon these research contributions by applying data analysis and clustering techniques to detect hotspots and provide a preventive approach for improving women's safety in public spaces.

B4. EXISTING SYSTEM

At present, several systems and applications have been developed to improve women's safety in public spaces. These systems are primarily designed to provide assistance during emergency situations and rely on mobile technology and location-based services.

Most existing solutions are mobile applications that offer features such as sending SOS alerts, sharing real-time location, and notifying emergency contacts. Applications like bSafe allow users to trigger alerts with a single action, enabling quick communication with trusted contacts. Similarly, My Safetipin provides safety scores for different locations based on factors such as lighting, crowd presence, and user feedback.

In addition to safety apps, mapping platforms such as Google Maps are widely used for navigation and location tracking. These platforms help users find routes and nearby places but are not specifically designed for safety analysis.

While these systems are useful, they have several limitations:

1. **Reactive Nature —** most existing systems respond only after an incident has occurred. They provide emergency assistance but do not help in preventing incidents.
2. **Dependence on User Interaction —** users must manually trigger alerts during emergencies, which may not always be possible in critical situations.
3. **Lack of Data Analysis —** existing systems do not analyze historical incident data to identify patterns or high-risk areas.
4. **Limited Predictive Capability —** these applications do not provide insights into where incidents are likely to occur in the future.
5. **Dependence on Internet Connectivity —** many features require continuous internet access, which may not be available in all situations.
6. **No Support for Urban Planning —** current systems do not provide insights that can help authorities in improving infrastructure or planning safer environments.

Due to these limitations, there is a need for a system that not only provides emergency support but also focuses on analyzing data and identifying unsafe areas in advance. This gap is addressed in the proposed system by introducing a preventive, data-driven approach.

B5. PROPOSED SYSTEM

The proposed system is designed to provide a data-driven and preventive approach to improve women's safety in public spaces. Unlike existing systems that mainly focus on emergency response, this system aims to analyze incident data and identify high-risk areas in advance.

The system collects data related to incidents such as geographic location (latitude and longitude), time of occurrence, and type of incident. This data is then preprocessed to remove inconsistencies and prepare it for analysis. By organizing the data effectively, the system ensures accurate and meaningful results.

To identify unsafe areas, the system applies clustering techniques such as K-Means Clustering. This technique groups similar data points based on their geographic proximity, allowing the system to detect regions where incidents are highly concentrated. These regions are identified as hotspots, indicating areas with higher risk.

The identified hotspots are then visualized using mapping and graphical techniques. A Heatmap is generated to represent the density of incidents using color intensity, where high-density areas indicate higher risk. This visualization makes it easier for users to understand and interpret complex data.

The system also incorporates spatial analysis concepts from Geographic Information Systems to study the geographic distribution of incidents. By analyzing both spatial and temporal patterns, the system can provide insights into when and where incidents are more likely to occur.

One of the key advantages of the proposed system is its ability to support preventive decision-making. The insights generated can help individuals choose safer routes and assist authorities in improving safety infrastructure such as lighting, surveillance, and patrolling. This makes the system useful not only for personal safety but also for urban planning.

Overall, the proposed system shifts the focus from reactive safety measures to a proactive approach by identifying risk areas in advance and enabling better planning for safer public environments.

B6. MODULES IDENTIFIED

The proposed system is divided into several modules to ensure a structured and systematic approach to data processing, analysis, and visualization. Each module performs a specific function and contributes to the overall working of the system.

The modules identified in this project are as follows:

B6.1 Data Collection Module

This module is responsible for gathering incident data related to women harassment. The data may include information such as location (latitude and longitude), time of occurrence, and type of incident. The data can be collected from publicly available datasets, reports, or simulated sources for analysis.

B6.2 Data Preprocessing Module

The collected data may contain missing values, errors, or inconsistencies. This module cleans and organizes the data by removing duplicates, handling missing values, and converting it into a structured format suitable for analysis. Proper preprocessing ensures accuracy in the results.

B6.3 Analysis Module

This module analyzes the processed data to identify patterns and trends. It examines factors such as frequency of incidents, time-based variations, and location-based distribution. This helps in understanding how incidents occur across different regions and time periods.

B6.4 Clustering Module

The clustering module applies algorithms such as K-Means Clustering to group similar data points based on geographic proximity. This helps in identifying high-risk areas (hotspots) where incidents are concentrated.

B6.5 Visualization Module

This module is responsible for presenting the results in an understandable format. It uses maps and graphical tools to display hotspots. Techniques like Heatmap are used to visually represent unsafe areas with different intensity levels.

B6.6 Reporting Module

The final module generates outputs in the form of reports or summaries. These outputs provide insights into identified hotspots and patterns, which can be used by users and authorities for decision-making and safety improvements.

B7. HARDWARE AND SOFTWARE REQUIREMENTS

The successful implementation of the proposed system requires a combination of basic hardware and software tools. The system is designed to be lightweight and does not require high-end infrastructure.

B7.1 Hardware Requirements

The project can be developed and executed on a standard computing system. The hardware specifications are summarized in Table 7.1 below.

Table B7.1: Hardware Requirements

Sl. No.	Component	Specification
1	Computer / Laptop	Standard desktop or laptop system
2	RAM	Minimum 4 GB (8 GB recommended)
3	Storage	At least 10 GB free disk space
4	Processor	Intel i3 or higher
5	Operating System	Windows / Linux / macOS

B7.2 Software Requirements

The system uses a combination of programming tools, libraries, and spatial technologies to perform data analysis and visualization. The complete software stack is listed in Table 7.2.

Table B7.2:Software Requirements and Tools Used

Sl. No.	Tool / Technology	Purpose
1	Python	Primary programming language for data analysis and visualization
2	Pandas	Data manipulation, cleaning and handling structured datasets
3	NumPy	Numerical computations and large-array handling
4	Matplotlib	Generating charts and graphs to visualize patterns
5	Folium	Interactive map visualization of incident locations and hotspots
6	K-Means Clustering	Grouping incident locations to detect hotspots
7	GIS Concepts	Spatial analysis of geographic distribution of incidents
8	Google Maps	Map visualization and geographic reference

Sl. No.	Tool / Technology	Purpose
9	Jupyter Notebook	Interactive environment for code, analysis and documentation
10	Visual Studio Code	Code editor for development and debugging
11	QGIS (Optional)	Advanced spatial visualization and professional GIS outputs
12	Git / GitHub	Version control, backup and collaboration

B7.3 Summary

The combination of Python, data analysis libraries, machine learning techniques, and spatial tools enables the system to analyze incident data efficiently and visualize high-risk areas. The inclusion of GIS concepts strengthens the project's relevance to spatial analysis and urban planning.

B8. SDG ALIGNMENT

This project is closely aligned with the global initiative of the United Nations under Sustainable Development Goal 5 (SDG 5): Gender Equality, which focuses on achieving gender equality and empowering all women and girls.

One of the key aspects of SDG 5 is addressing violence against women in both public and private spaces. The project specifically relates to:

3. **Target 5.2 —** eliminate all forms of violence against women and girls, including harassment and unsafe conditions in public areas such as streets, transportation systems, and marketplaces.
4. **Indicator 5.2.2 —** measures the proportion of women and girls subjected to sexual violence by persons other than an intimate partner, based on location and context of occurrence.

The proposed system contributes to achieving this goal by adopting a preventive and data-driven approach. Instead of only responding to incidents, the system analyzes historical data to identify high-risk areas (hotspots) where such incidents are more likely to occur. By using spatial analysis and visualization techniques, the system provides valuable insights into unsafe locations.

These insights can be utilized by authorities and urban planners to take necessary actions such as improving street lighting, increasing police patrolling, installing surveillance systems, and designing safer public infrastructure. Additionally, individuals can use this information to make safer travel decisions and avoid high-risk areas.

By supporting both awareness and preventive planning, the project contributes to creating safer public spaces and promoting gender equality. Thus, it aligns with the broader vision of SDG 5 by helping reduce violence and ensuring a safer environment for women.

B9. WORK BREAKDOWN STRUCTURE

The development of the project is planned in a systematic manner by dividing it into multiple phases. Each phase focuses on a specific task, ensuring proper organization, timely completion, and efficient management of the project.

B9.1 Phase 1: Problem Definition and Research (Week 1)

In this phase, the problem is clearly identified and studied in detail. Background research is conducted to understand women's safety issues, existing systems, and gaps. Relevant technologies such as spatial analysis and data analytics are also explored.

B9.2 Phase 2: Data Collection (Week 2)

In this stage, data related to incidents is collected from available sources. The data includes information such as location, time, and type of incident. This forms the foundation for further analysis.

B9.3 Phase 3: Data Preprocessing (Week 3)

The collected data is cleaned and organized by removing errors, handling missing values, and converting it into a structured format. This step ensures that the data is suitable for accurate analysis.

B9.4 Phase 4: Data Analysis and Clustering (Week 4)

In this phase, the processed data is analyzed to identify patterns. Clustering techniques are applied to group similar data points and detect high-risk areas (hotspots).

B9.5 Phase 5: Visualization (Week 5)

The results obtained from analysis are visualized using maps and graphical tools. Techniques such as heatmaps are used to clearly represent unsafe areas and make the data easy to interpret.

B9.6 Phase 6: Testing and Result Evaluation (Week 6)

The system is tested to ensure correctness and efficiency. The results are evaluated to verify whether the identified hotspots and patterns are meaningful and accurate.

B9.7 Phase 7: Documentation and Presentation (Week 7)

In the final phase, the complete project is documented, including all sections such as introduction, methodology, and results. A presentation is prepared to explain the project clearly during evaluation.

B9.8 Project Schedule (Gantt Chart)

The Gantt chart shown in Fig. 9.1 illustrates the planned schedule for each phase of the project across a seven-week timeline. The shaded cells indicate the duration during which each task is actively being performed.

Phase	Task	W1	W2	W3	W4	W5	W6	W7
P1	Problem Definition & Research							
P2	Data Collection							
P3	Data Preprocessing							
P4	Data Analysis & Clustering							
P5	Visualization							
P6	Testing & Result Evaluation							
P7	Documentation & Presentation							

Fig. B9.1: Project Schedule – Gantt Chart

- Active State

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